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Chapter 6

Execute Day End Functions



Introduction To Execute Day End Functions

Housekeeping's Day-End Functions perform two major activities.

1. The saving of libraries and folders to save files for subsequent saving to media (see the Save to Media chapter).
2. Activities to make best use of your AS/400; deleting libraries not defined to Housekeeping, compressing libraries, and re-organising the physical files contained within libraries.

The above activities should be performed as part of an overnight routine. To perform them on a regular basis it is recommended that they are run as part of Machine Manager's Auto Day-End.

Machine Manager is provided with Housekeeping's Day-End Functions as Standard Jobs (see the Auto Day-End Jobs Chapter of the Machine Manager Product Guide). This makes the setting up of these tasks very simple and ensures that they are run regularly without the need for manual intervention.

Before you initiate any of the Day-End Functions you should first make sure that you have set up Housekeeping's Control Details and Definitions (see Chapters Control Details and Definitions).

Libraries

This function performs the following activities:

- Saving libraries to save files.
- Deletes libraries tagged for deletion.
- Deletes libraries tagged for archiving (see User Libraries section). The deletion is not performed if a security copy was requested and has not yet been performed.
- Clears out work libraries (see Library Types section).
- Re-sizes libraries, and analyses when they last changed, if the library's save frequency indicates it should be saved.

A library will be saved to a save file specified in Housekeeping's Control Information (see Control Information section) if:

- The library is due for saving, that is, the number of days since the last save to media equals or exceeds the library's save frequency, and the library has changed since the last save to media. A library's save frequency is maintained via Library Definitions (see the Definitions Chapter).
- Or , the library is due for saving and the force save indicator is set to 1 (yes). The force save indicator is maintained via Library Definitions (see the Definitions Chapter).
- Or, the library has been flagged for deletion and a security save to media has been requested. See sections; Library Deletion, Deletion Requests, and Deletion Audit.
- Or, the library has been flagged to be archived. See section User Libraries which explains the way in which a library is flagged to be archived
- Or, the library has been flagged to force a save tonight. A library may have been flagged to force a save tonight by either the User Libraries function (see User Libraries section) or the Free Library Media function (see Free Library Media section).

Once the save to save file has been completed then the library may be saved to media (see Libraries section in the Save To Media chapter).

Libraries Not Saved

There are a number of situations where even though Housekeeping calculates that a library is due to be saved to a save file, the save is not performed.

- If the library is in use then Housekeeping will not attempt to save it to the save file.
- If the user who initiates the Day-End function has insufficient authority to perform the save.
- The disk overhead in auxiliary storage used by the save file generated may cause critical storage to be reached, or even passed. Before Housekeeping performs the saves to save file it calculates the potential storage needed for each library. If the total potential storage needed exceeds Housekeeping's critical storage threshold, then the save to save file will not occur.



Note: The data area HKDISKLMT holds the disk utilisation threshold at which saves to save files are to be terminated. This is initially set at 97%.



Note: Error messages from Housekeeping's Day-End functions are sent to a message queue called HKSAVERR.

Folders

This function performs the saving of those folders due to be saved to the save file specified in Housekeeping's Control Information (see Control Information section).

A folder will be saved to the save file if:

- The folder is due for saving, that is, the number of days since the last save to media equals or exceeds the folder's save frequency, and the folder has changed since the last save to media. A library's save frequency is maintained via Library Definitions (see Folders section in the Definitions chapter).
- Or, the folder is due for saving and the force save indicator is set to 1 (yes). The force save indicator is maintained via Library Definitions (see Folders section in the Definitions chapter).
- Or, the folder has been flagged to force a save tonight. Section Folders in the Operational Enquiries describes how a folder may be force saved.

Once the save to save file has completed the folder may be saved to media (see Archiving Libraries section).

Folders Not Saved

There are a number of situations where even though Housekeeping calculates that a folder is due to be saved the save is not performed.

- If the folder is in use then Housekeeping will not attempt to save it to the save file.
- If the user who initiates the Day-End function has insufficient authority to perform the save.
- The disk overhead in auxiliary storage used by the save file generated may cause critical storage to be reached, or even passed. Before Housekeeping performs the saves to save file it calculates the potential storage needed for each folder. If the total potential storage needed exceeds Housekeeping's critical storage threshold, then the save to save file will not occur.



Note: The data area HKDISKLMT holds the disk utilisation threshold at which saves to save files are to be terminated. This is initially set at 97%.



Note: Error messages from Housekeeping's Day-End functions are sent to a message queue called HKSAVERR.

Off Site Libraries

Each library may belong to a pre-defined group for the purpose of saving a number of libraries as a set in order to store at an off-site location. This group of libraries is known as an Off-Site Save Group.

The Library Definitions function (see Libraries section in the Definitions chapter) enables you to designate the off-site group to which a library belongs.

Only one off-site save group may be saved at any one time. The selection of a group of libraries to be saved is performed by the function Select Off-Site Group (see Select Off-Site Group section).



Note: Before you run this Day-End function to save off-site libraries you must select an off-site group to be saved.

The libraries are saved to a save file which can subsequently be saved directly to media (see Select Off-Site Group section).

Off Site Libraries Not Saved

There are a number of situations where even though Housekeeping calculates that a library is due to be saved to a save file, the save is not performed:

- If the library is in use then Housekeeping will not attempt to save it to the save file.
- If the user who initiates the Day-End function has insufficient authority to perform the save.
- The disk overhead in auxiliary storage used by the save file generated may cause critical storage to be reached, or even passed. Before Housekeeping performs the saves to save file it calculates the potential storage needed for each library. If the total potential storage needed exceeds Housekeeping's critical storage threshold, then the save to save file will not occur.



Note: The data area HKDISKLMT holds the disk utilisation threshold at which saves to save files are to be terminated. This is initially set at 97%.



Note: Error messages from Housekeeping's Day-End functions are sent to a message queue called HKSAVERR.

Select Off Site Group



Select option 4 within Day-End Function and press **Enter**.

The purpose of this function is to select the off-site group which will be used by the next run of the day-end function to save off-site libraries (see Off-Site Libraries section in the Execute Day-End Functions chapter).

The screenshot shows a terminal window with the following content:

```
HK150                               Housekeeping                               System: STUDB30A
Select Save Group

Enter the code of the Library Save Group to be saved by the next run of
the Auto-Backup Library Save Group routine.

If a Save Group has already been designated, it will be shown.

Press ENTER to update/confirm.

Save Group  __

F3=Exit  F4=Browse
```

Fields:

Save Group

Enter the two character code of the off-site save group you wish to save in the next overnight run.

Delete Unknown Libraries

This function literally and physically deletes from your IBM AS/400 all of the libraries which are not known to Housekeeping.

Before running this function you should ensure that you either have all of the libraries you wish not to be deleted either defined to Housekeeping (see Libraries section in the Definitions topic), or protected by the Protected Libraries function (see Protected Libraries section).



Warning: It is important that you check that libraries you wish to keep are either defined to Housekeeping, or protected by the Protected Libraries Function.

Re Organise Libraries

This function examines each library defined to Housekeeping and determines whether or not that library is flagged to have its files re-organised (see Libraries section in the Definitions topic).

For each of the libraries selected the physical files contained within them will be re-organised.

The re-organisation of physical files frees up the space occupied by deleted records and in some instances can lead to significant savings in disk space.

Compress Libraries

This function examines each library defined to Housekeeping and determines whether or not that library is flagged to be compressed (see Libraries section in the Definitions topic).

For each of the libraries selected the programs contained within those libraries will have their observable information removed.

Removing the observability of program objects can lead to significant savings in disk space without affecting the performance of the program (though it does restrict the information provided by program dumps).



Note: Removing the observable information from program objects restricts the information provided by program dumps. Therefore it is advisable not to remove the observable information from programs until they are stable. All JBA Software Products Limited products are shipped with the observable information removed.

Re Size Libraries

This function examines each library defined to Housekeeping (see Libraries section in the Definitions topic) and determines its size. The library size information produced is used in a number of areas within Housekeeping:

- A libraries size history may be viewed in bar chart form (see Library Size History and Libraries Total Size History sections).
- Housekeeping's documentation provides a report Libraries Not Changed Since Given Date (see Not Changed Since Given Date section) which utilises information provided by the library re-sizing routines.